

Hive Metastore -> Unity Catalog

Classic Compute -> Serverless

Mount Point -> Volumes / External Location

Live Keyword - Remove that and use 3 level namespace.

Lakeflow Connect (Standard Connectors)

Database (Managed Connector)

azure sql database

- orders
- customers

CT (change tracking)

CDC (change data capture)

SQL Server

Databricks

connection to the database - credentials (unity catalog)

get the data from database to your volume (staging area) - gateway

from volume (Staging area) data is ingested in your delta tables - pipeline

Gateway (classic compute) - will keep running

Ingestion pipeline (batch) - serverless (volume and bring it to delta lake)

if you have not increased the quota for your cpu cores then you will get error.

EDV4 - driver (4 cores)

Standard FS - worker (4 cores)

## Lakeflow Jobs

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Lakeflow connect (ingestion)

Lakeflow declarative pipelines (ETL)

Lakeflow Jobs (orchestration)

Airflow, ADF

The challenge with the 3rd party orchestration tools,  
not unified with Delta lake, integration becomes hard.

what is a job here?

It consists of one or more tasks..

task - is a single unit of work within a job. It can execute specific workloads such as notebook, scripts, sql query etc...

task

- ingesting customers from raw customer table
- ingesting orders from a folder

think job as a container which holds the entire workflow.

when you create a job you can set specific configurations for each particular task.

- defining the path to the code
- setting parameters
- enable notifications
- configure the retry policies.

Lakeflow jobs

- consists of one or more tasks (notebooks, dlt, sql files, python scripts)
- control flows can be established between tasks (sequential, parallel, conditional, for each)
- jobs support different triggers (scheduled, file arrival trigger, table trigger, continuous trigger, manual trigger)

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Medallion architecture

Bronze -> Silver -> Gold

Orders

Customers

In SQL server we have a customers table -> Lakeflow connect -> Delta table  
(Customers Raw table)

orders (table in sql server) -> a third party tool is bringing incremental orders and  
putting in a csv file within a folder in ADLS gen2.

abfss://dataset@lakeflowstorageaccount.dfs.core.windows.net/orders

so 2 sources of data

customers (ingest\_schema.customers)

orders (csv file in orders folder)

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orders raw

customer raw is ready

customers\_bronze

customers bronze (sql file)

customer silver cleaned (notebook)

customer silver enriched (notebook)

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Gold Layer

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| city      | total_sales | state     |
|-----------|-------------|-----------|
| bangalore | 10000       | Karnataka |
| hyderabad | 5000        | Telangana |

large\_city

true  
false

total\_sales > 10000

